

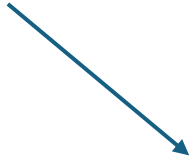
Iteration:

Loss:

1	oshcime urjhrnin	3.42
2	mshcine lrnhrrnin	2.85
3	machnie learnrnn	1.67
4	machine learninn	0.42
5	machine learning	0.00

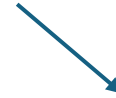
Dave Braun
May 6, 2025

Humans are flexible



Machines are rigid

```
print('Hello world.')
```



```
>>> Hello world
```

Goal: Perform as well as possible in a soccer match

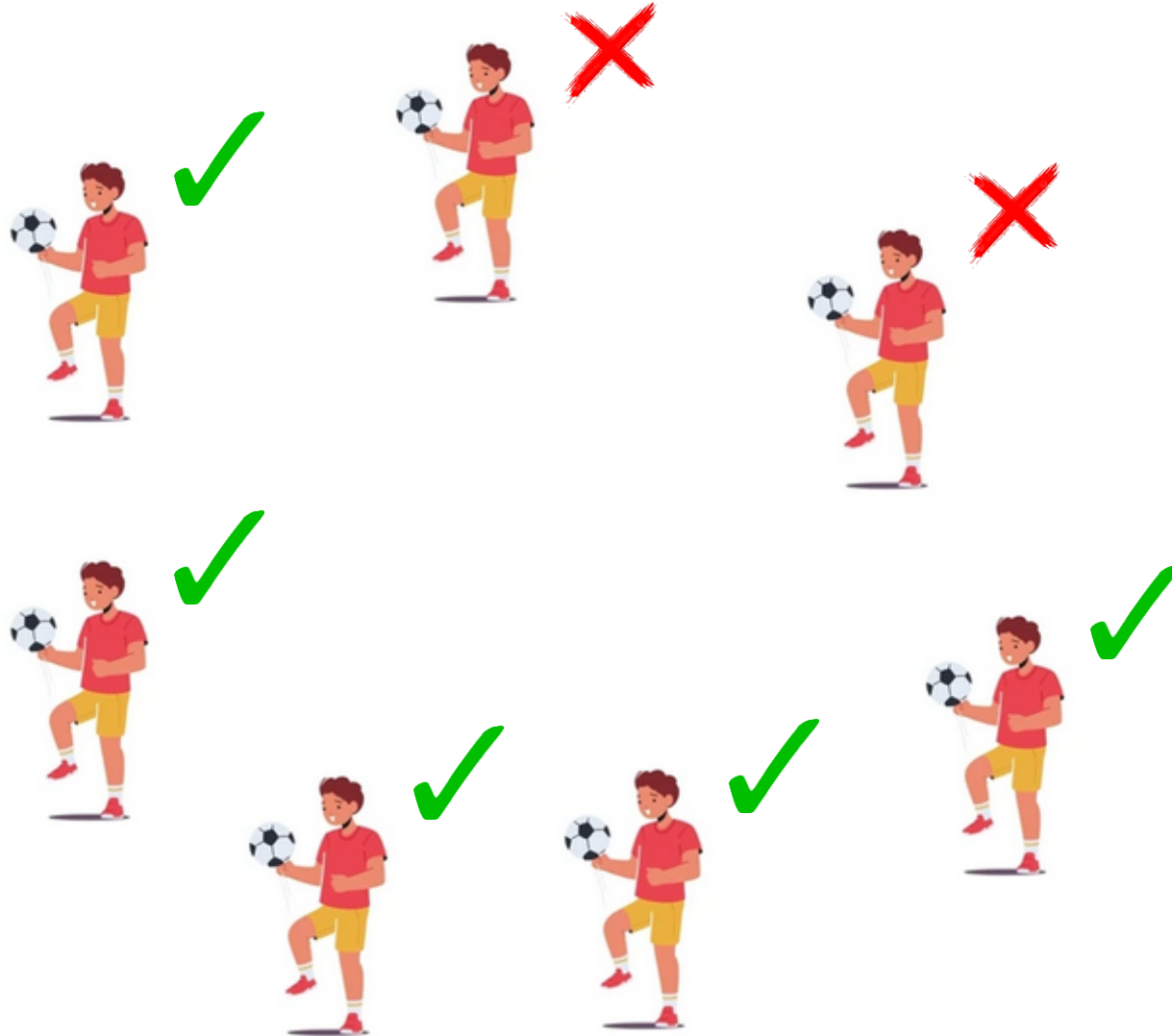
Step 1: Practice many skills and get feedback

Step 2: Perform! Put practice into action.



Goal: Perform as well as possible in a soccer match

Step 1: Practice many skills and get feedback



Step 2: Perform! Put practice into action.

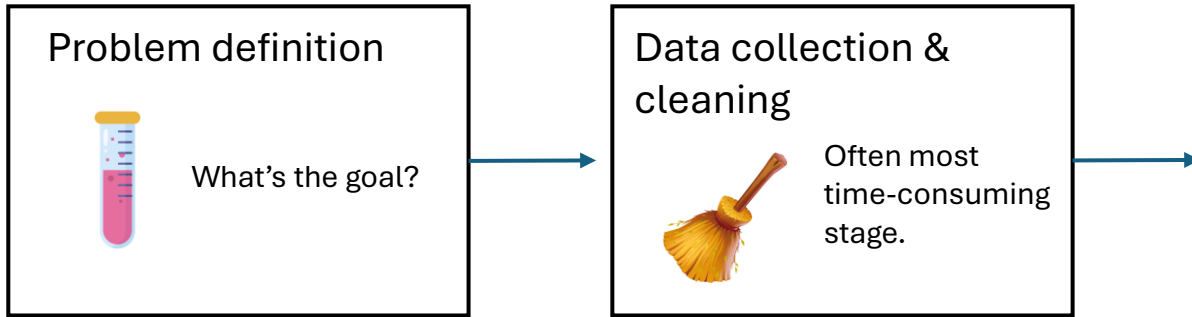


Too rigid!

How to make machines flexible learners?

Two important components:

1. Learn from previous experience (ie, data)
2. Generalize beyond previous experience (ie, be flexible!)



Machine learning workflow

How to make machines flexible learners?

Goal: Predict depression scores as accurately as possible given health data

id	age	sleep	exercise
500	54	5.8	8
501	36	7.3	2
502	22	8.4	5
...	...	...	...



id	depression
500	5.2
501	3.1
502	6.3
...	...



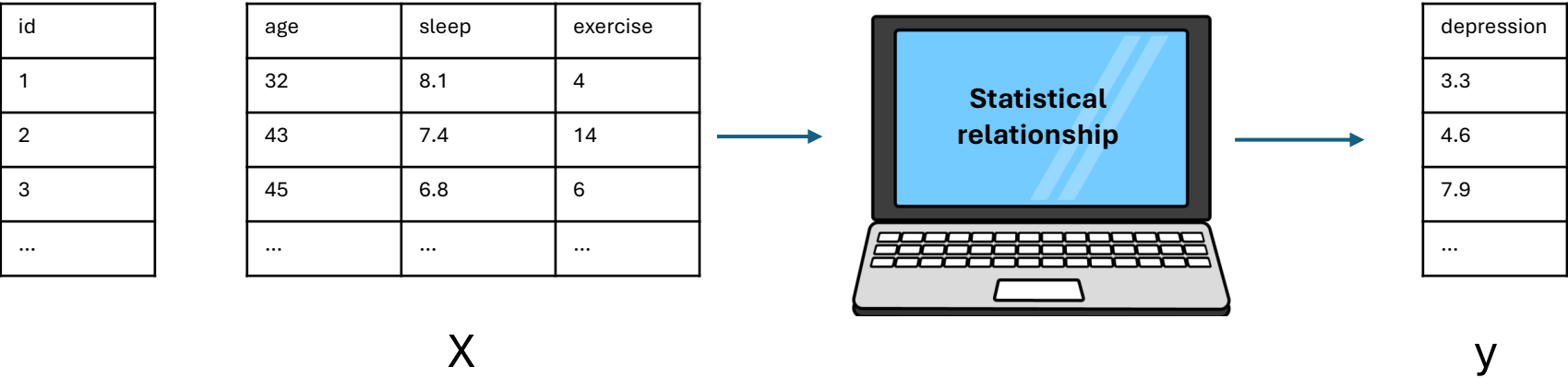
Basic approach: Learn a statistical relationship

Imagine we knew depression levels for some people

id	age	sleep	exercise	depression
1	32	8.1	4	3.3
2	43	7.4	14	4.6
3	45	6.8	6	7.9
...	...	...	...	...

Step 1: Learn from previous experience

We could learn a statistical relationship between health data and depression scores



Basic approach: Use the statistical relationship to make guesses about new data

We learned a statistical relationship between health data and depression scores from existing data

age	sleep	exercise
32	8.1	4
43	7.4	14
45	6.8	6
...	...	...

X



depression
3.3
4.6
7.9
...

y

Step 1: Learn from previous experience

Step 2: Generalize beyond previous experience

We use that statistical relationship to make a guess about depression levels for new individuals

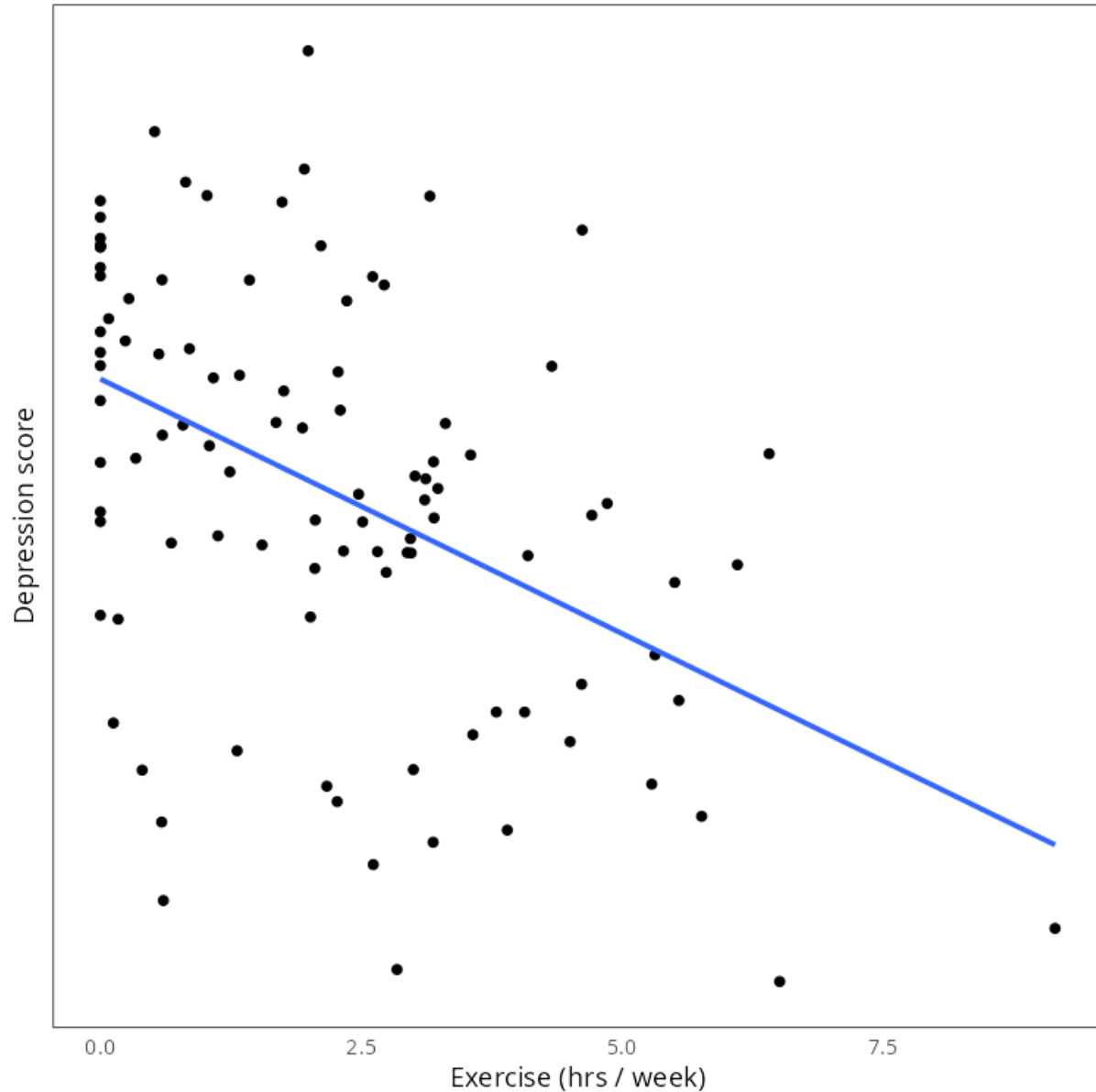
Id	age	sleep	exercise	depression
500	54	5.8	8	?
501	36	7.3	2	?
502	22	8.4	5	?
...	...	...	...	...



$\hat{y}$

depression
5.2
3.1
6.3
...

Step 1: Learn a statistical relationship between health data and depression scores



id	exercise	depression
1	8	3.3
2	2	4.6
3	5	7.9
...	...	...

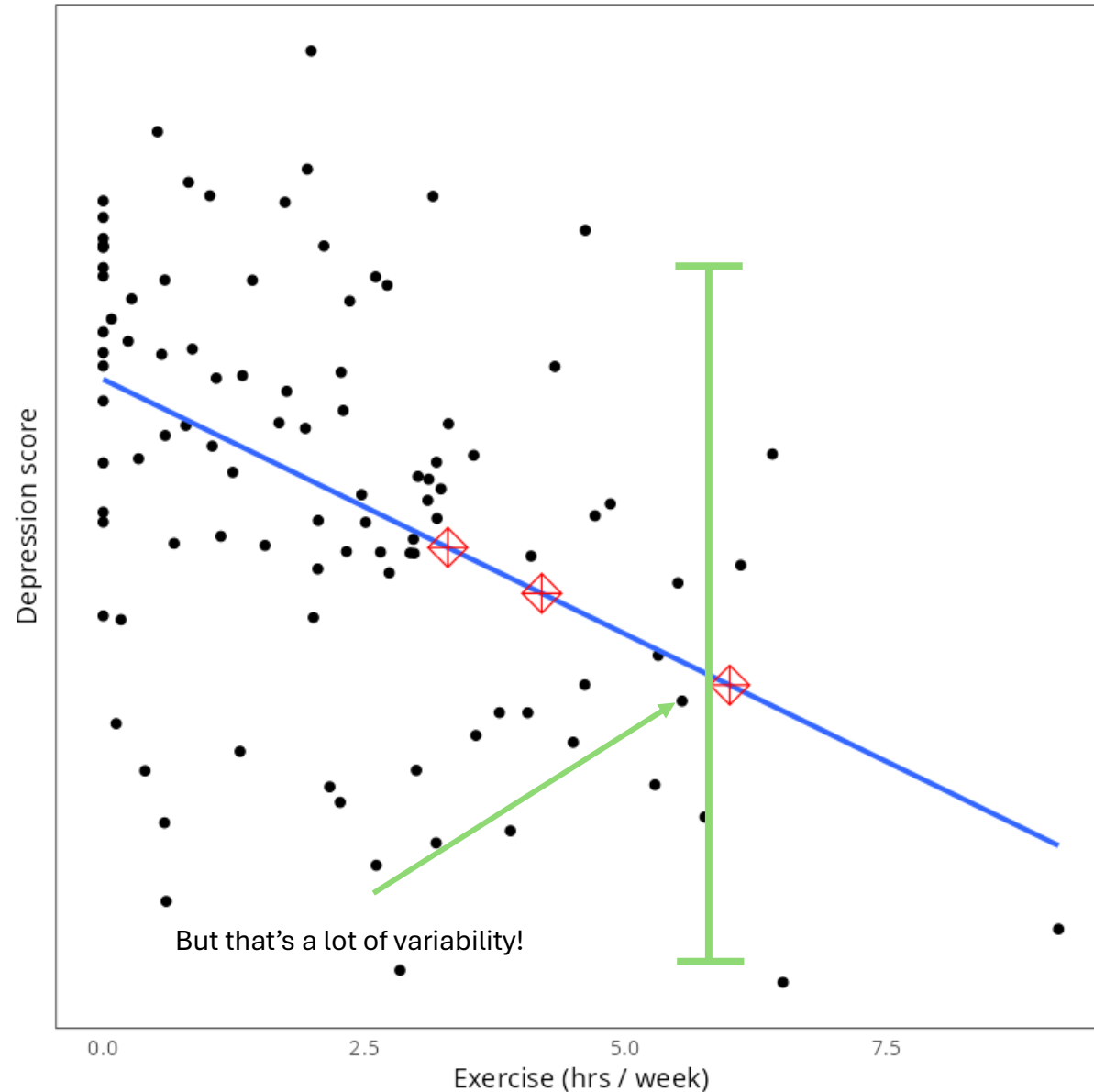
Original data

$$depression = b_0 + exercise \cdot b_1$$

Statistical
relationship

The machine “learns” values of b_0 and b_1 to make a line that best fits the data

Step 2: Use that statistical relationship to infer depression levels for new individuals



$$\text{depression} = b_0 + \text{exercise} \cdot b_1$$

Statistical
relationship

id	exercise	depression
501	3.3	?
502	6	?
503	4.2	?
...	...	...

New
individuals

If we know b_0 , b_1 , and exercise hours per week, we can make an inference about depression scores!

What do b_0 and b_1 tell us about the nature of the relationship between exercise and depression?

Is b_1 “significant”?

Will the relationship between exercise and depression generalize to the population?

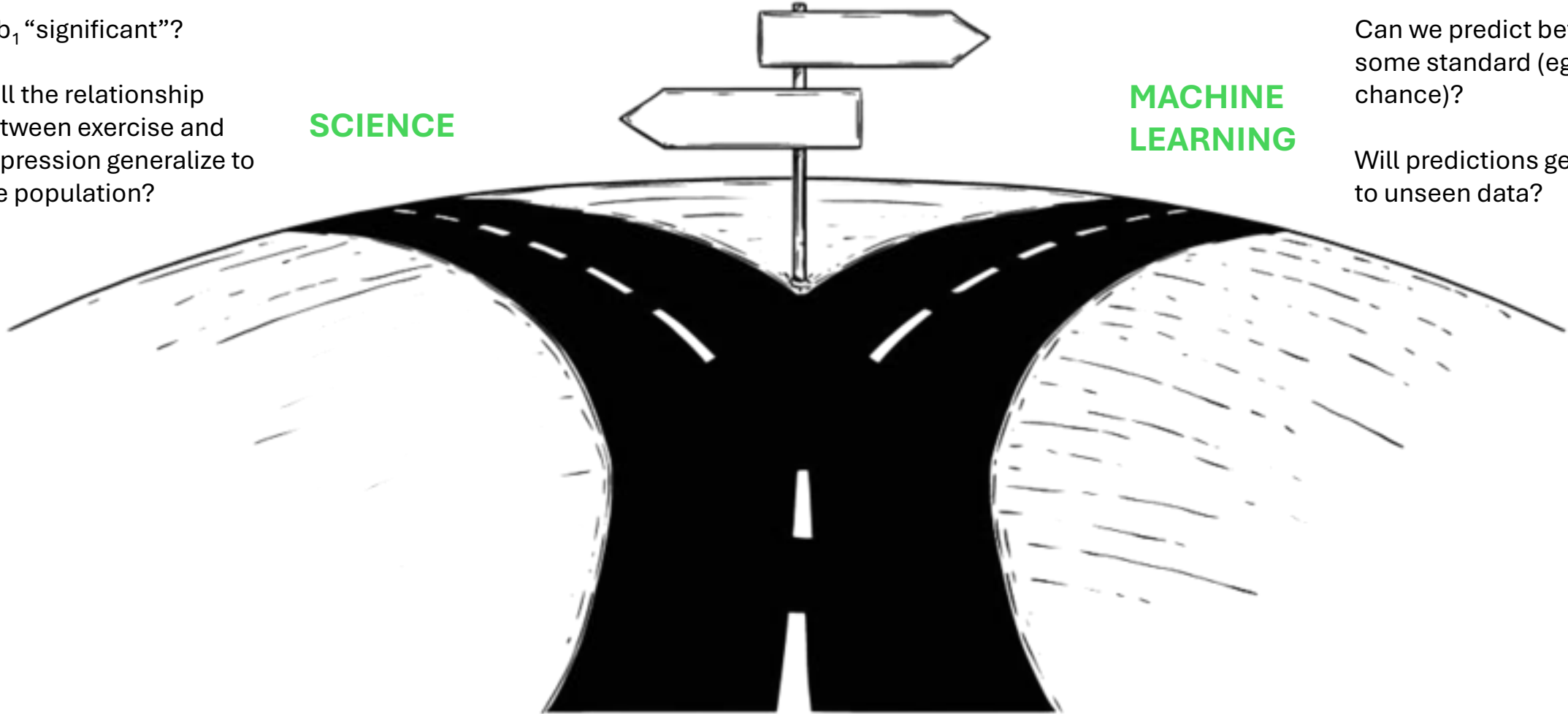
SCIENCE

MACHINE
LEARNING

How can we predict depression scores as accurately as possible given health data?

Can we predict better than some standard (eg, chance)?

Will predictions generalize to unseen data?

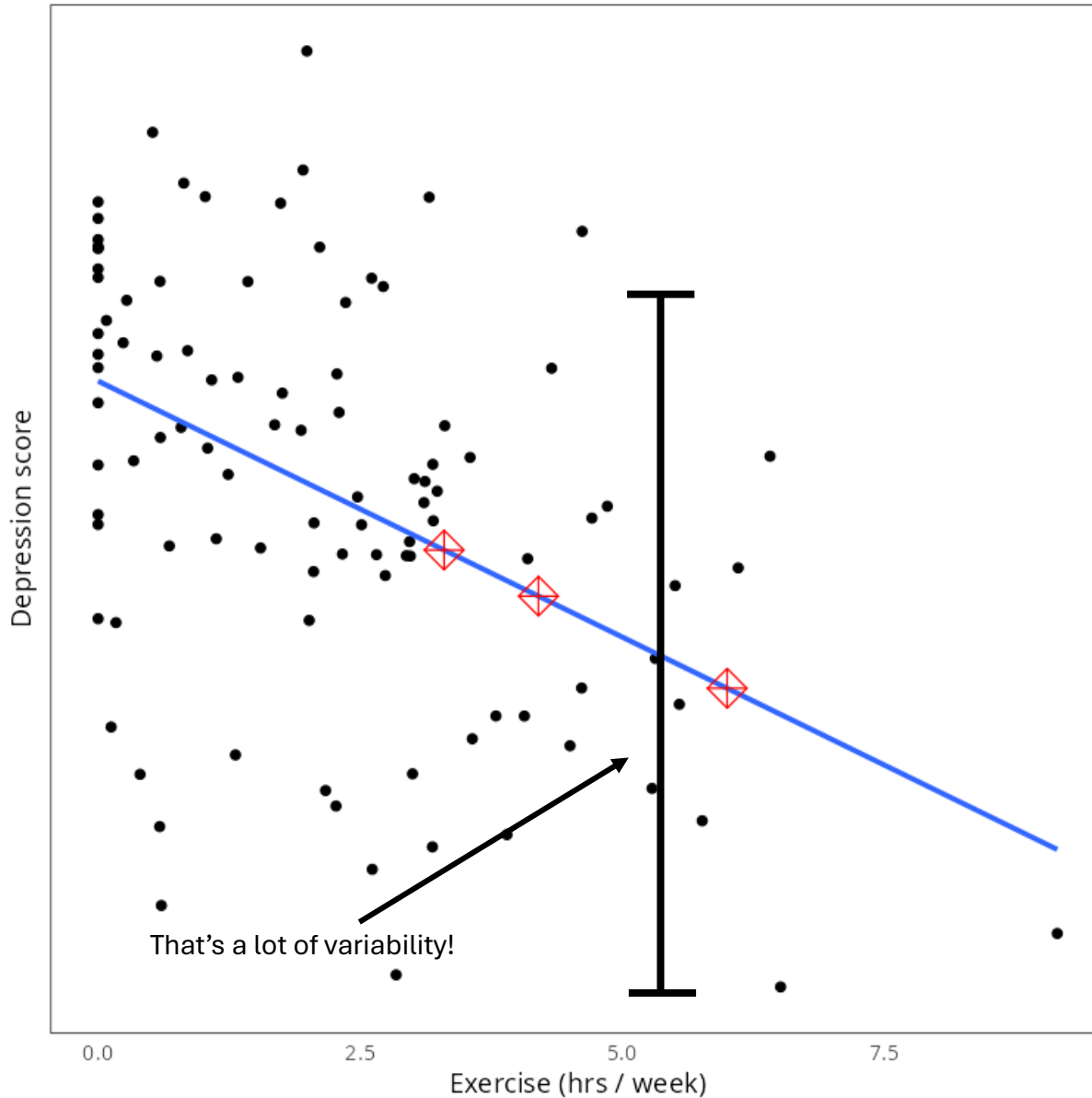


What we're learning: Statistics and machine learning rely on the same tools but apply them in different ways



Academic scientists
deciphering data science
buzzwords

Gatekeeping data
scientists



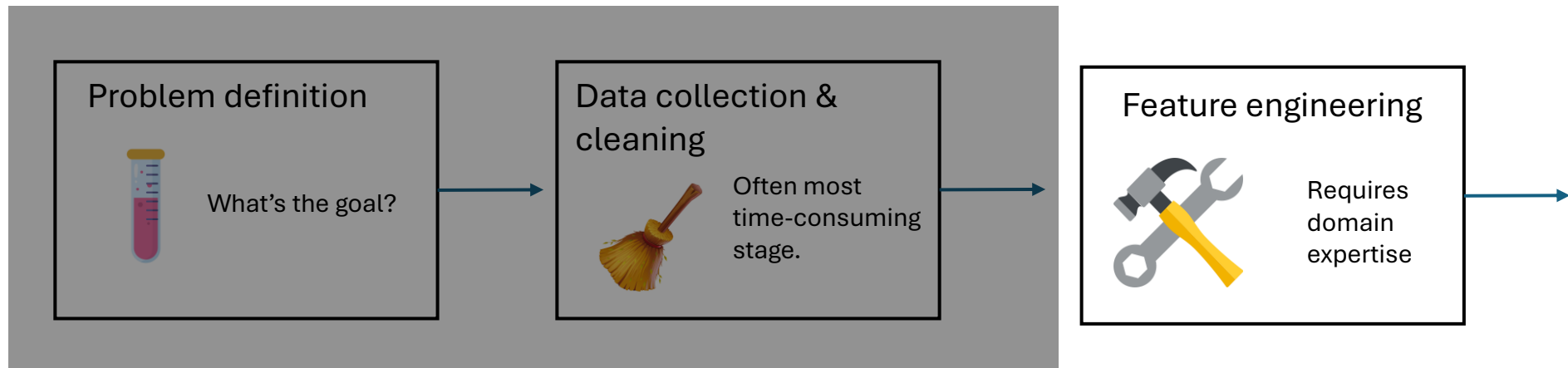
$$\text{depression} = b_0 + \text{exercise} \cdot b_1$$

id	exercise	depression
501	3.3	?
502	6	?
503	4.2	?
...	...	...

Goal: Predict depression scores as accurately as possible given health data

1. Learn from previous experience (ie, data)
2. Generalize beyond previous experience (ie, be flexible!)

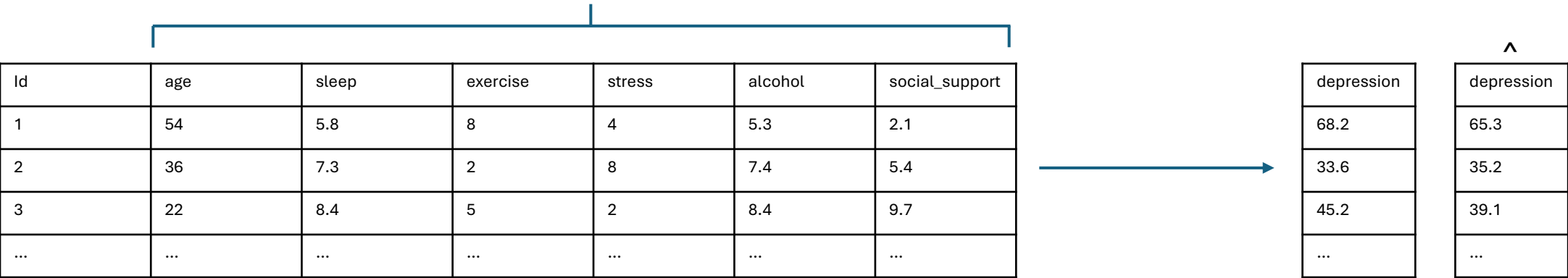
How can we do better?



Machine learning workflow

Making the dataset “wider” will lead to more accurate predictions

(as long as the features are actually associated with your target and are independent from each other)



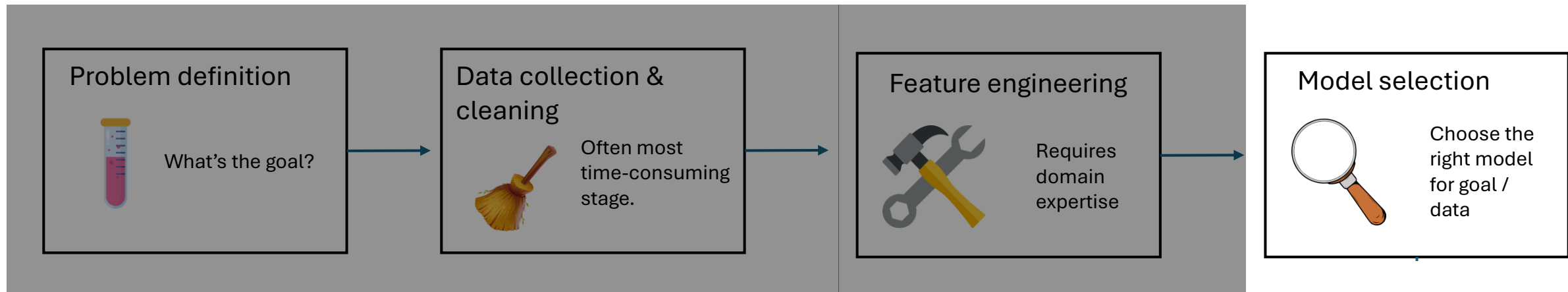
“feature”

Can no longer visualize this in a scatterplot 🌈

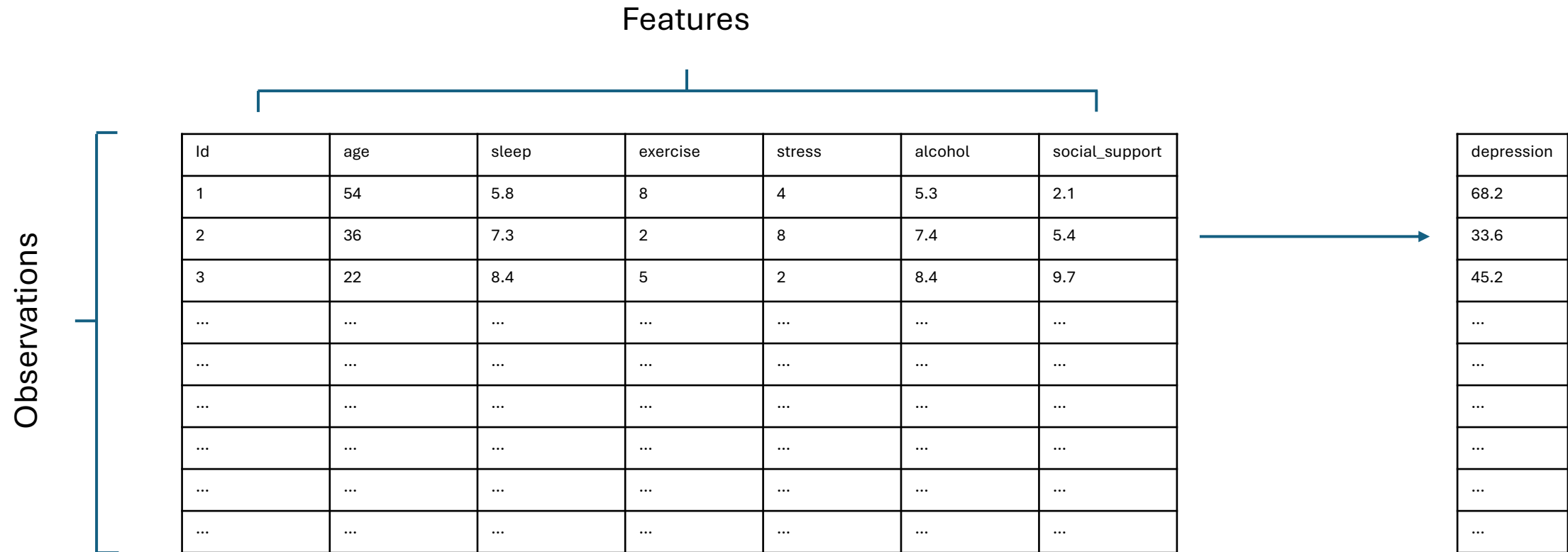
$$depression = b_0 + b_1 \cdot age + b_2 \cdot sleep + b_3 \cdot exercise + \dots$$



Step 1: Learn from previous experience



Machine learning workflow



Few features,
many observations

Many features,
many observations

Linear / logistic
regression



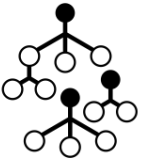
Naïve Bayes



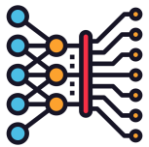
K-Nearest neighbors



Random forests



Deep learning

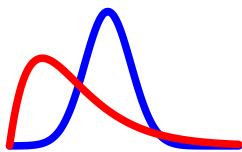


More observations

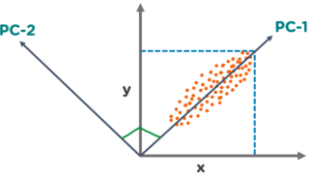
Linear / logistic
regression



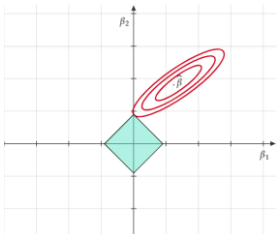
Bayes with priors



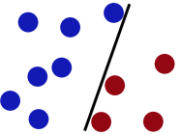
PCA + regression



Ridge regression

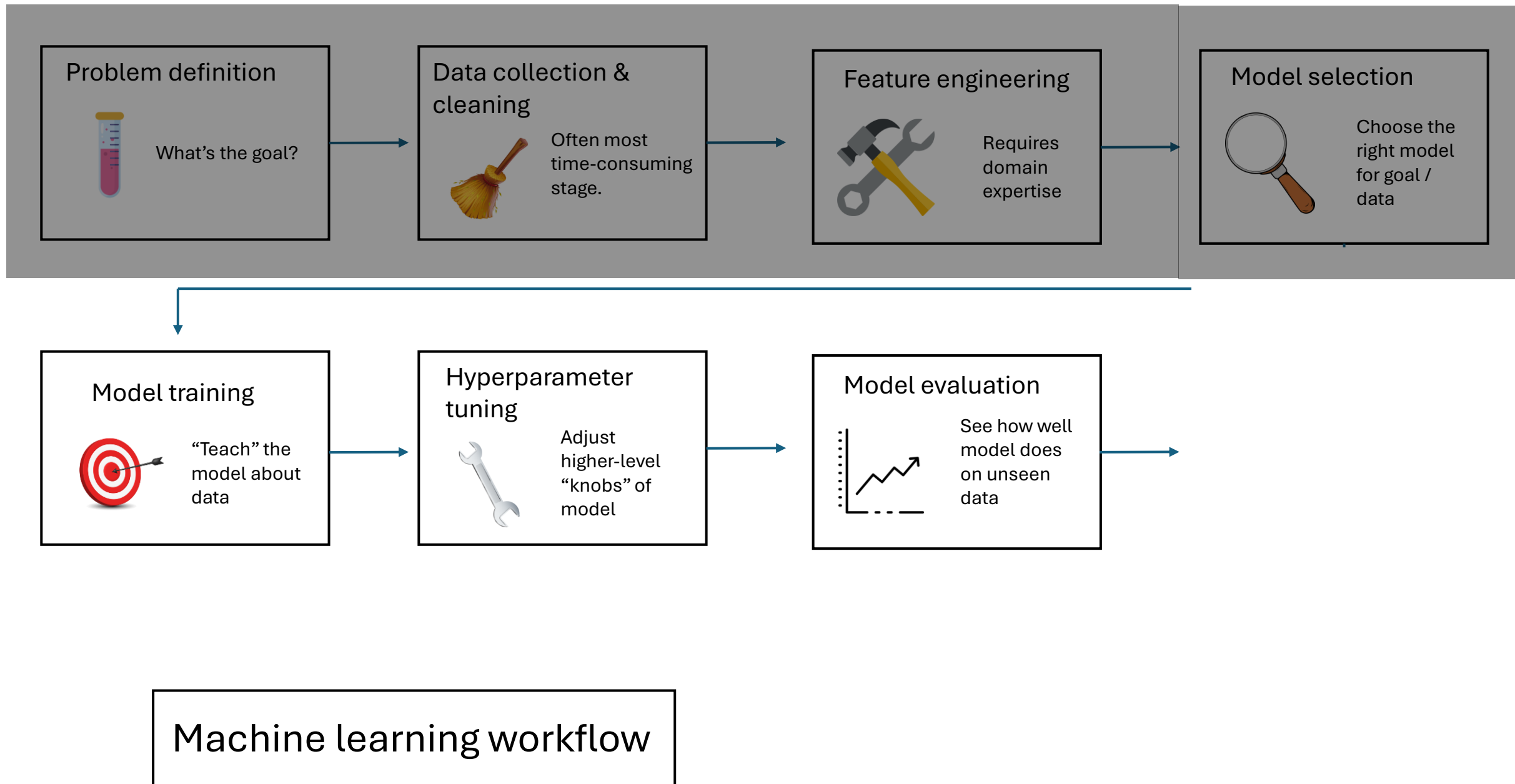


Support vector
machine



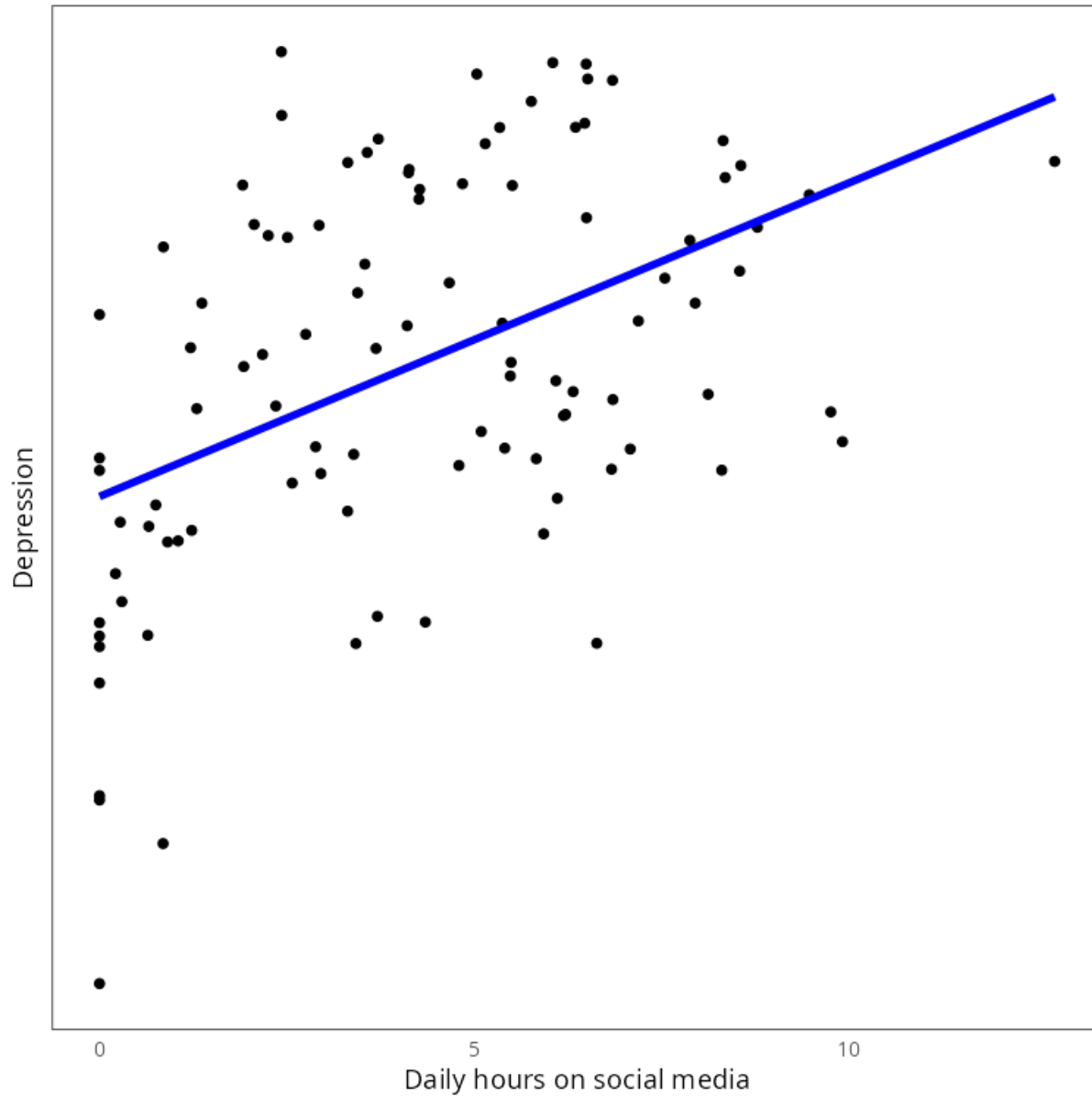
Few features,
few observations

Many features,
few observations



Goal: Predict depression scores as accurately as possible given health data

1. Learn from previous experience (ie, data)
2. Generalize beyond previous experience (ie, be flexible!)

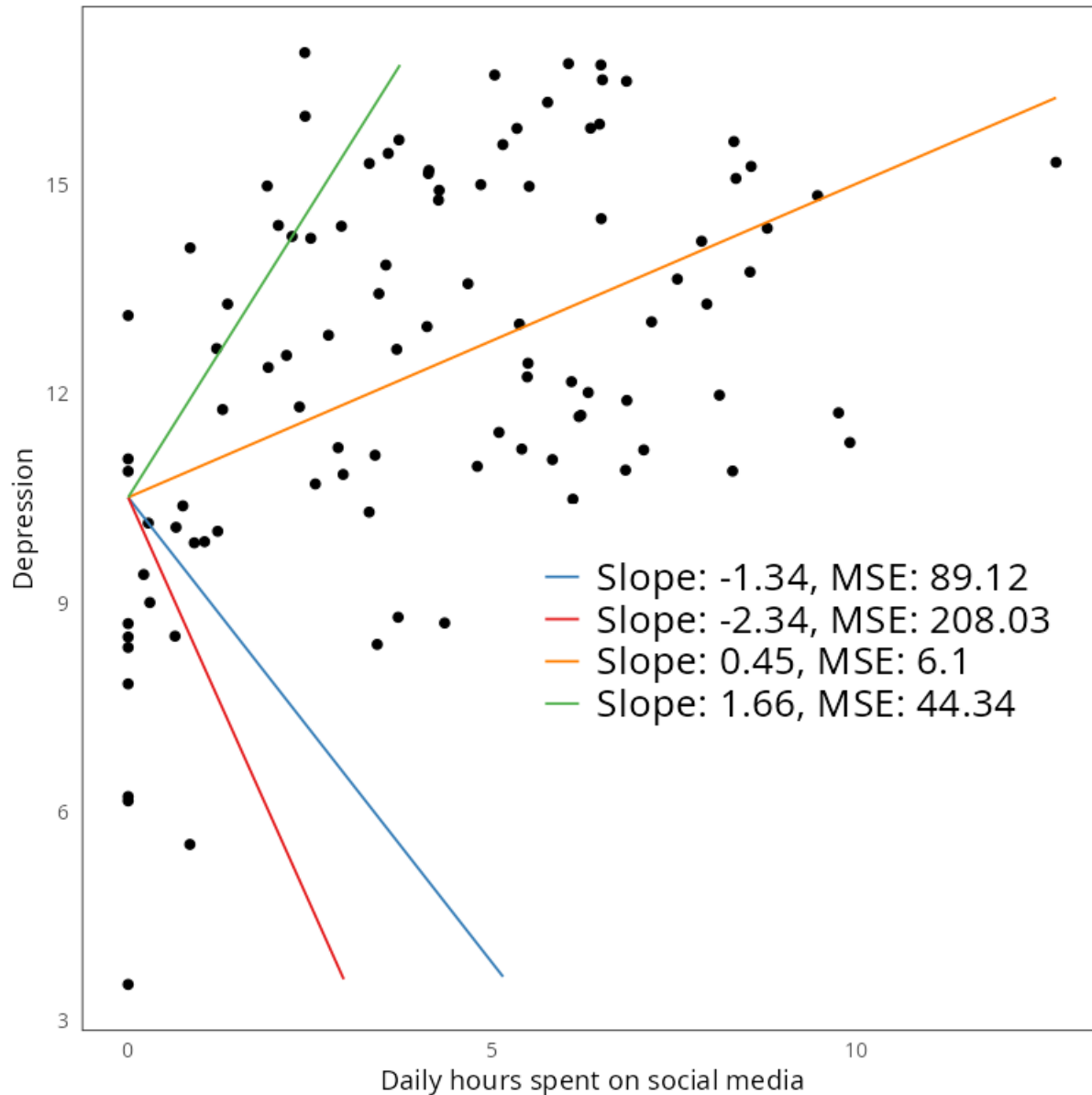


Goal: Predict depression scores as accurately as possible given health data

1. Learn from previous experience (ie, data)
2. Generalize beyond previous experience (ie, be flexible!)

$$depression = b_0 + b_1 \cdot social$$

How does a model “learn” b_0 and b_1 ?

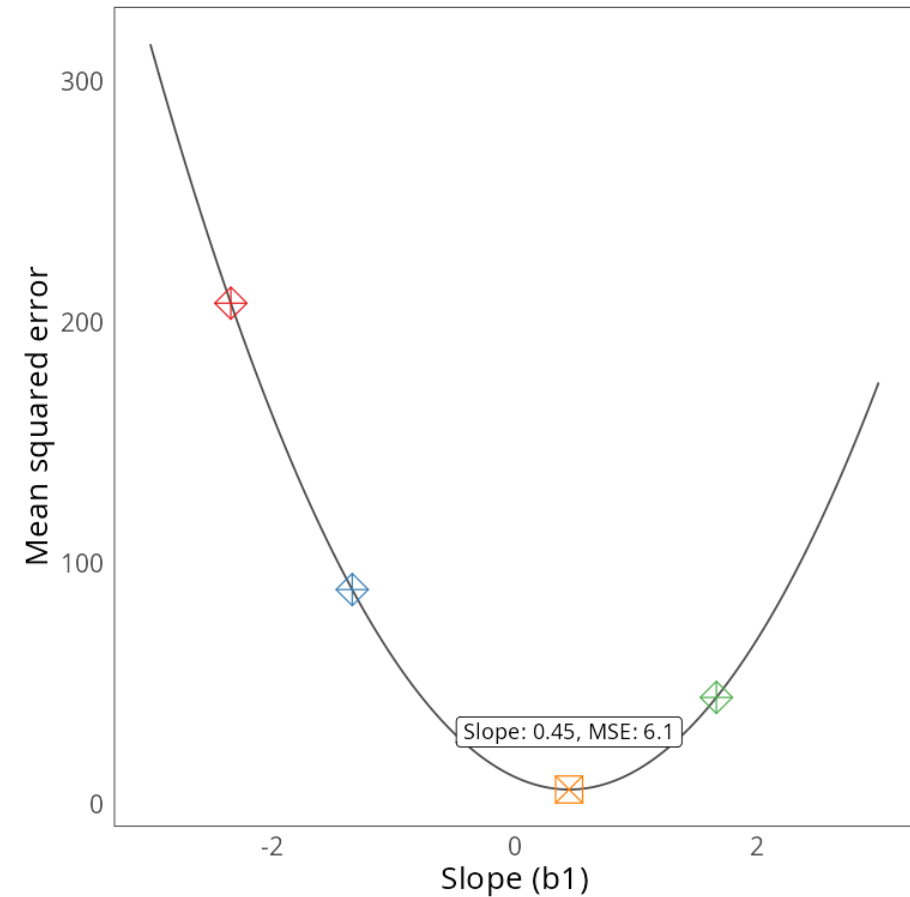


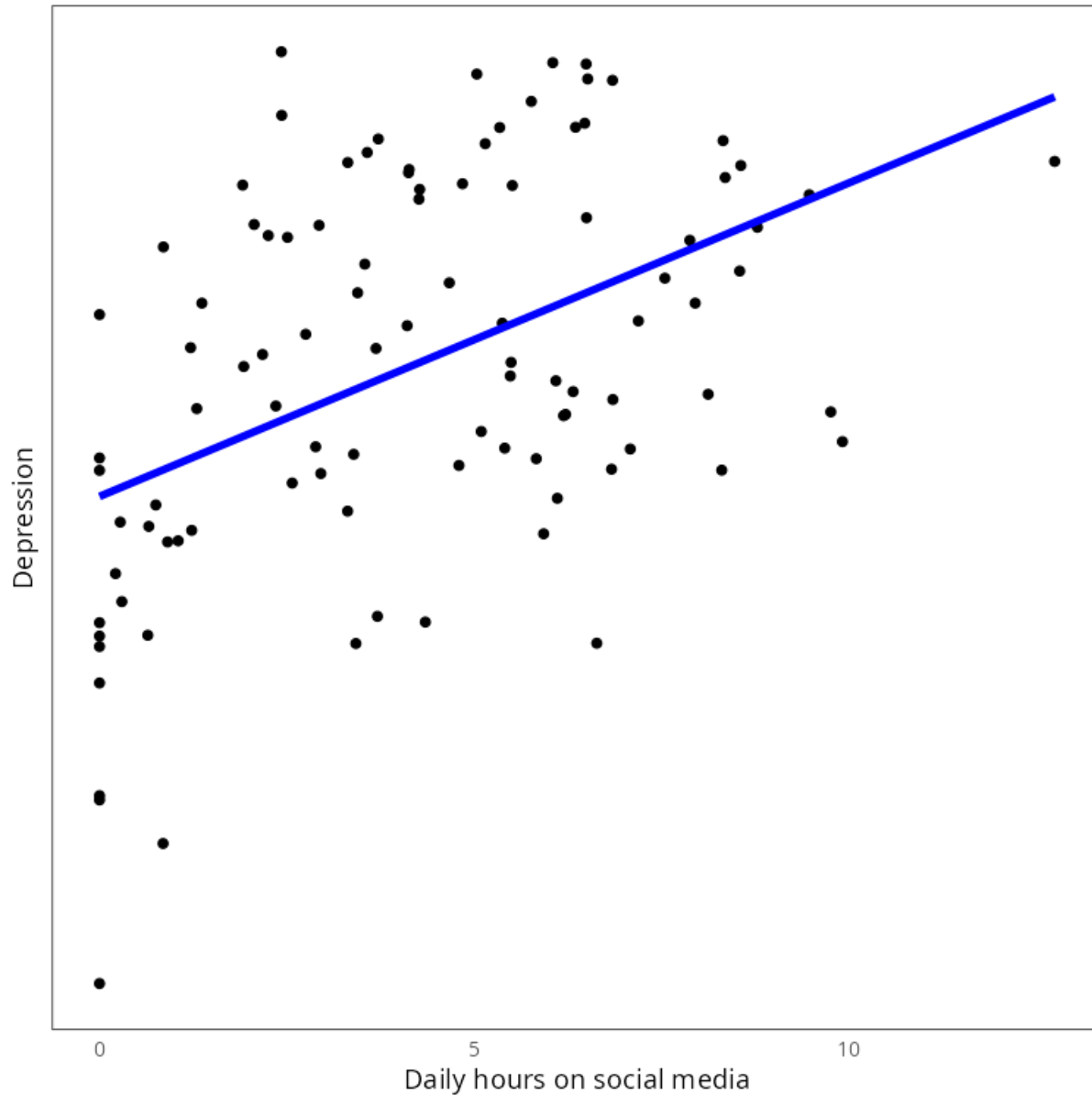
Goal: Predict depression scores as accurately as possible given health data

1. Learn from previous experience (ie, data)
2. Generalize beyond previous experience (ie, be flexible!)

$$\text{depression} = b_0 + b_1 \cdot \text{social}$$

How does a model “learn” b_1 ?





Goal: Predict depression scores as accurately as possible given health data

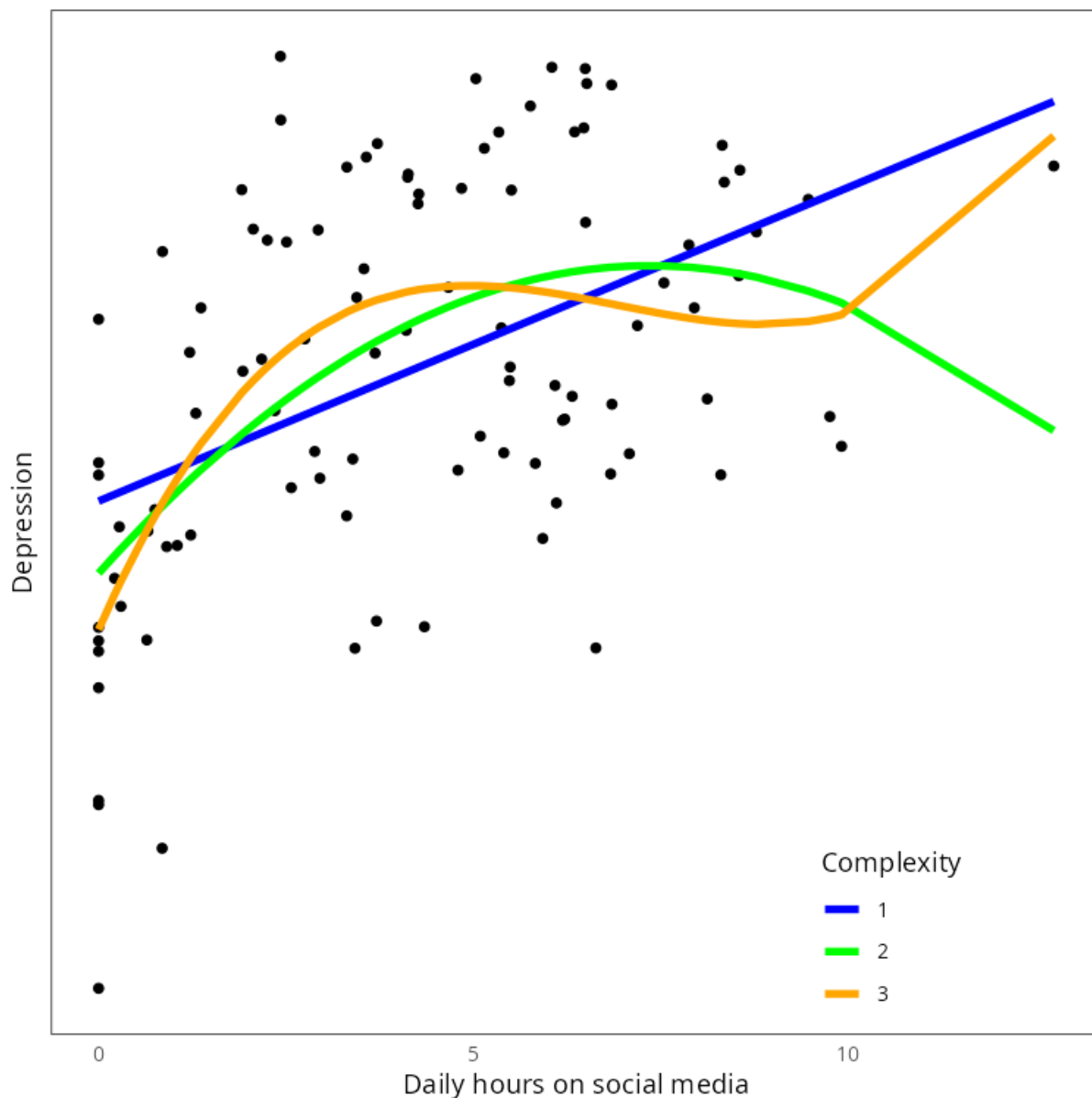
1. Learn from previous experience (ie, data)
2. Generalize beyond previous experience (ie, be flexible!)

$$depression = b_0 + b_1 \cdot social$$

$$MSE = 3.39$$

Two ways to optimize **Step 1:**

1. Increase number of features
2. Increase model complexity



Goal: Predict depression scores as accurately as possible given health data

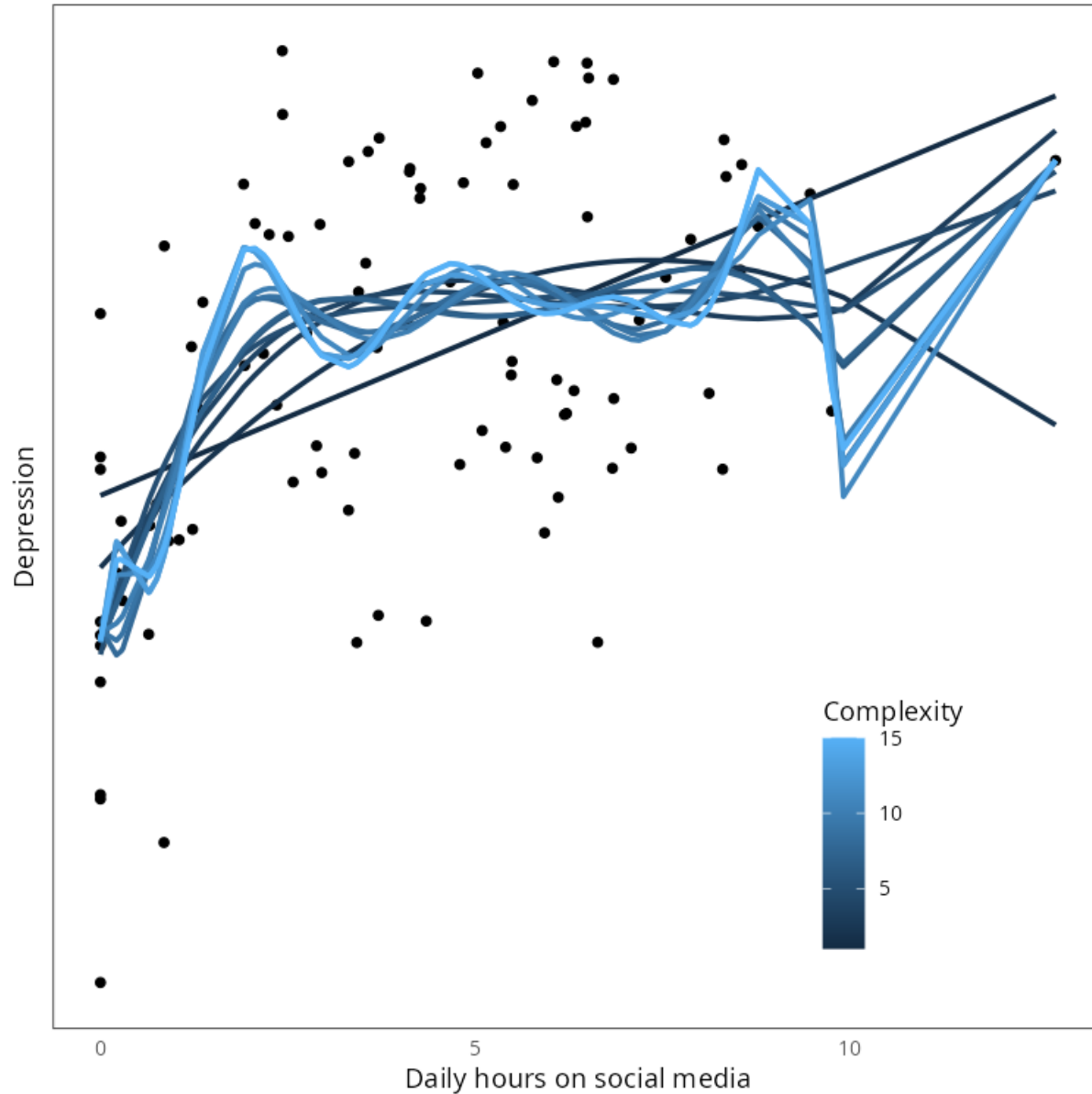
1. Learn from previous experience (ie, data)
2. Generalize beyond previous experience (ie, be flexible!)

Degree (complexity)	Mean squared error
1	3.39
2	3.290

$$depression = b_0 + b_1 \cdot social$$

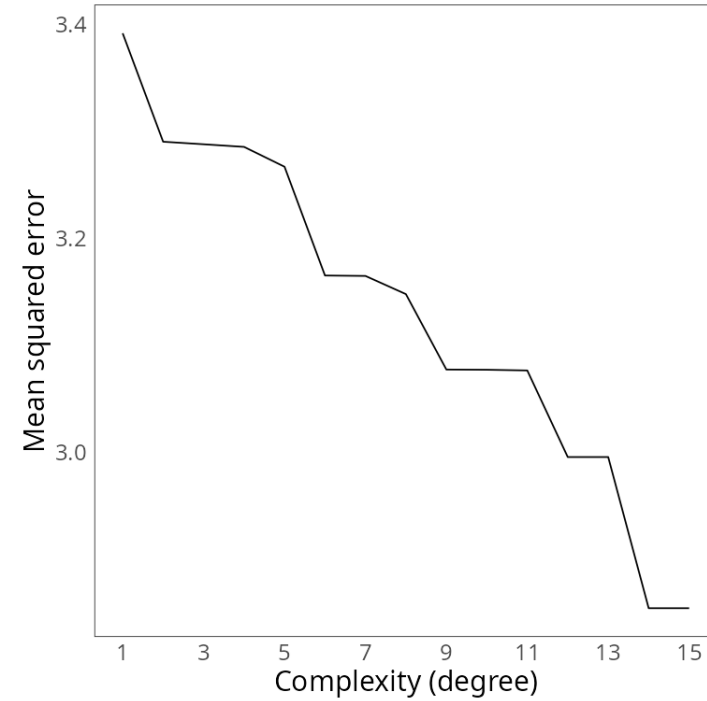
$$depression = b_0 + b_2 \cdot social^2 + b_1 \cdot social$$

$$depression = b_0 + b_3 \cdot social^3 + b_2 \cdot social^2 + b_1 \cdot social$$



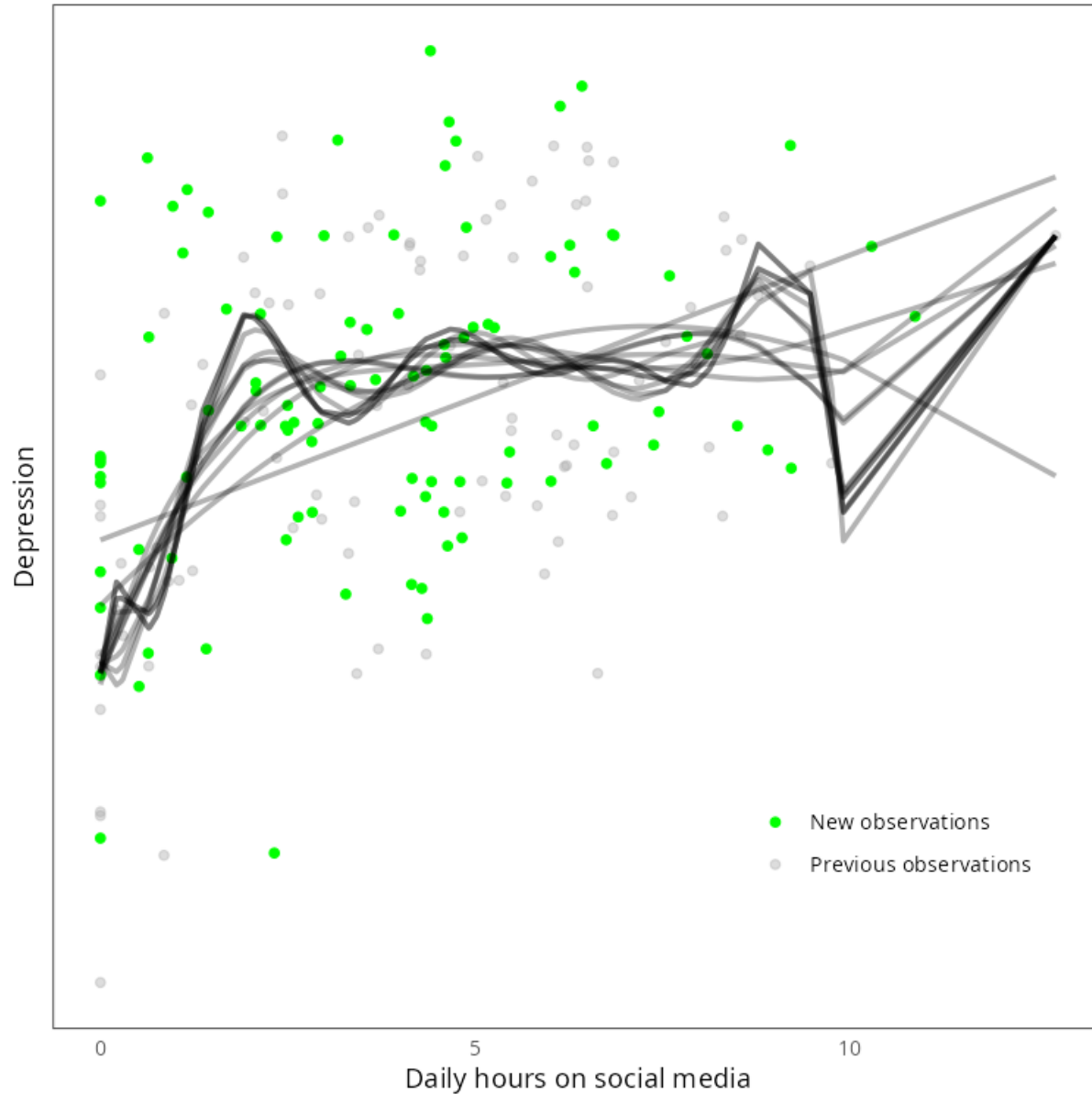
Goal: Predict depression scores as accurately as possible given health data

1. Learn from previous experience (ie, data)
2. Generalize beyond previous experience (ie, be flexible!)



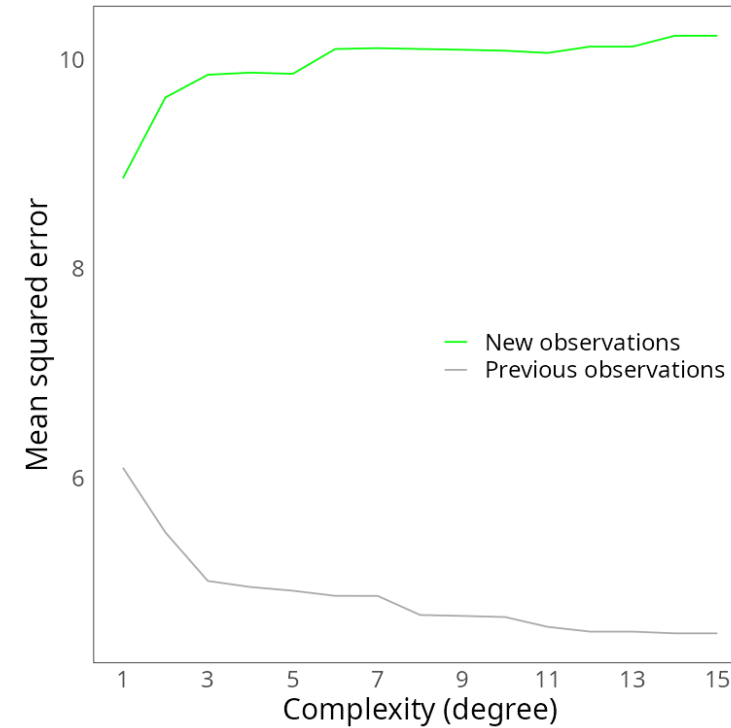
Learn from previous experience (ie, data)





Goal: Predict depression scores as accurately as possible given health data

1. Learn from previous experience (ie, data)
2. Generalize beyond previous experience (ie, be flexible!)



Learn from previous experience (ie, data)



Generalize beyond previous experience



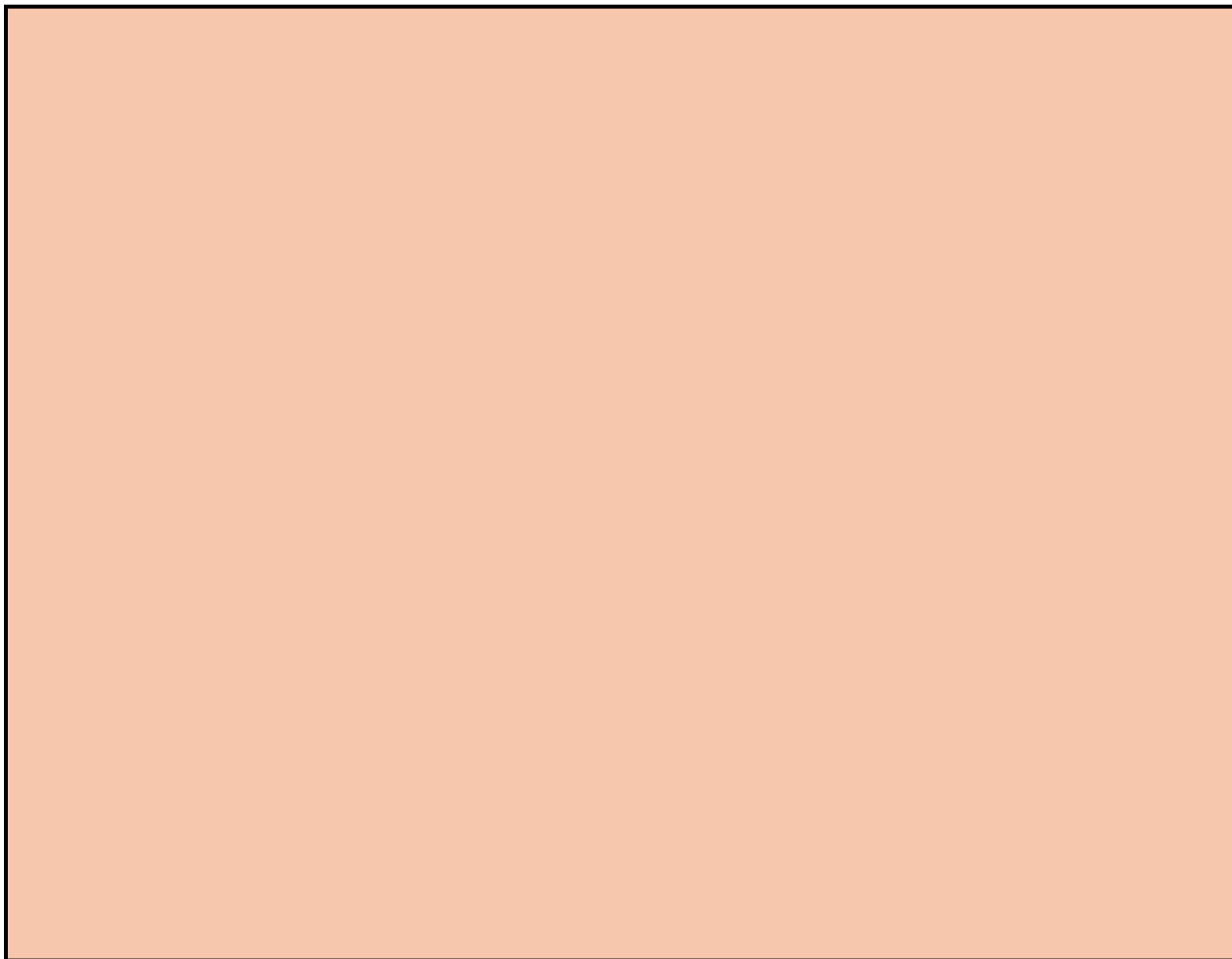
X

age	sleep	exercise	stress	alcohol	social_support
54	5.8	8	4	5.3	2.1
36	7.3	2	8	7.4	5.4
22	8.4	5	2	8.4	9.7
...	...	...	...	...	...
...	...	...	...	...	...
...	...	...	...	...	...
...	...	...	...	...	...
...	...	...	...	...	...
...	...	...	...	...	...

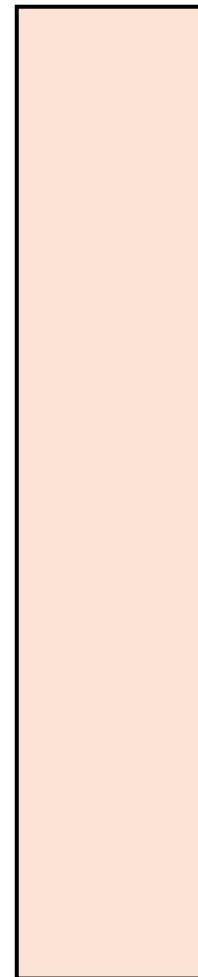
y

depression
68.2
33.6
45.2
...
...
...
...
...
...

x



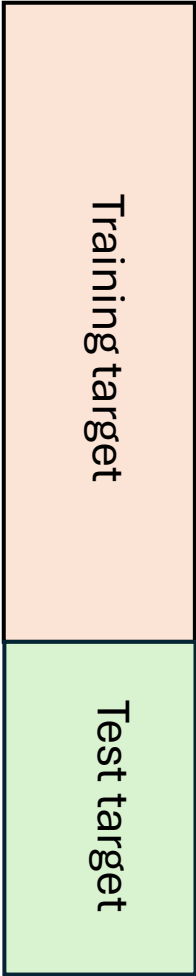
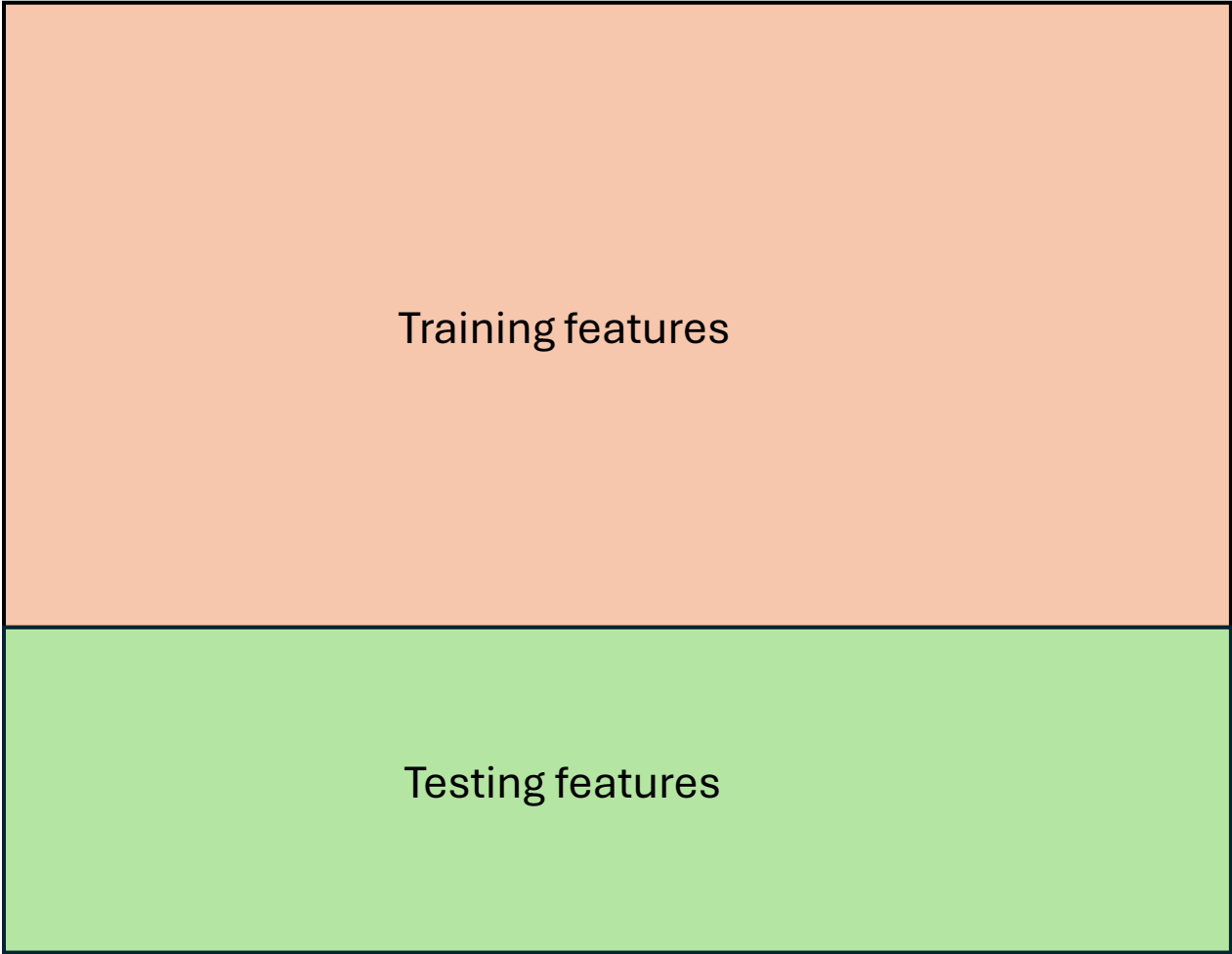
y



Get the model set up (“fitted”) on only this data first

X

y

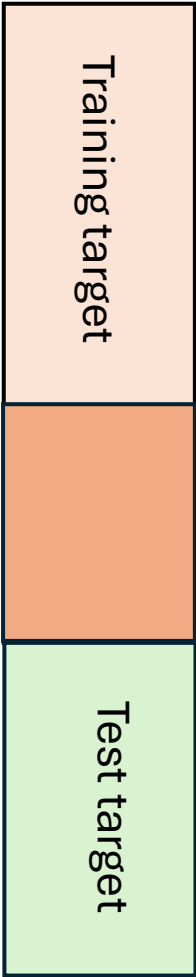
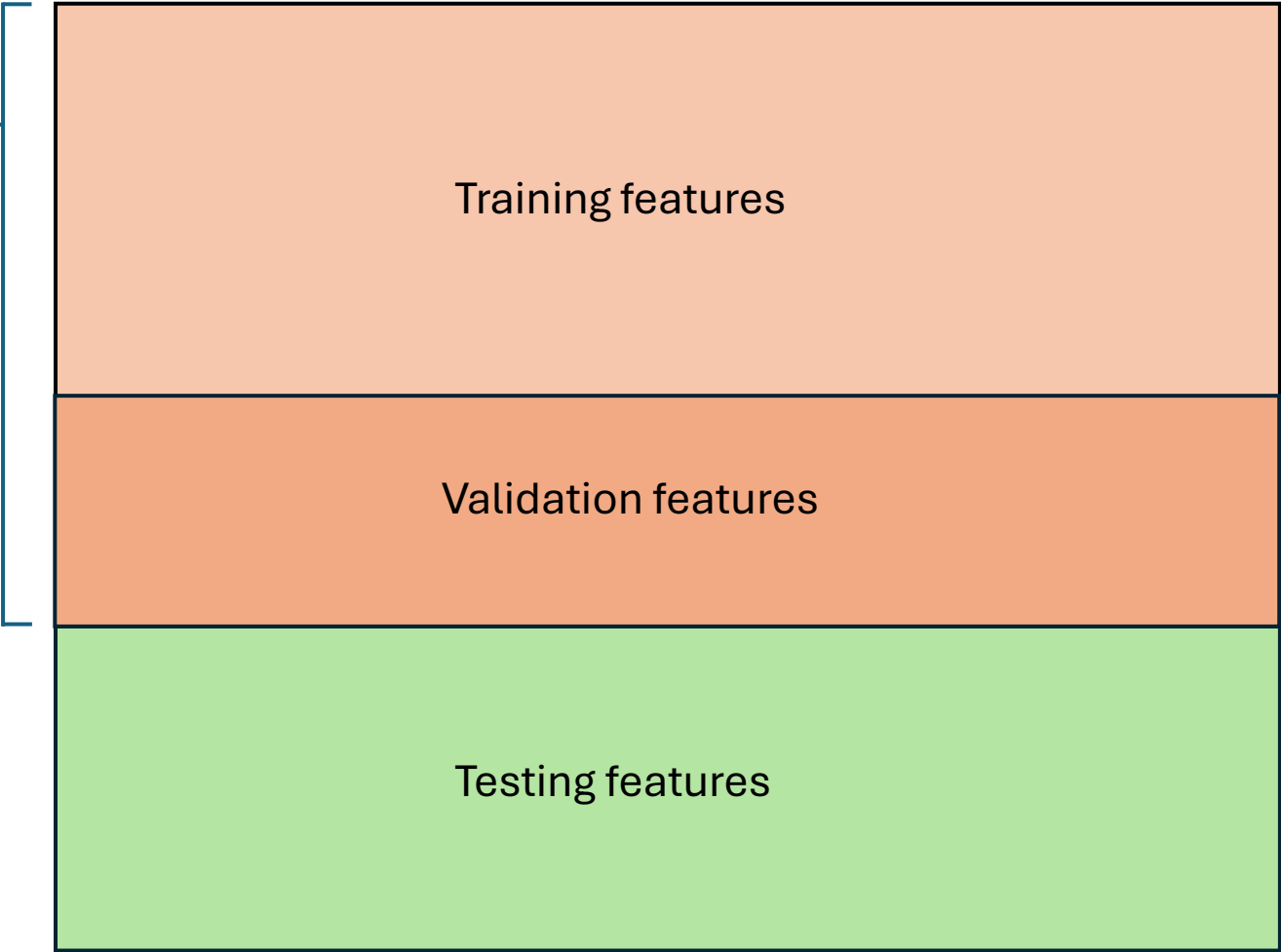


Then see how well the model can generalize by testing it on this data that the model has never seen

Can “tune” the model by training and checking multiple times on this data, without overfitting to testing data

X

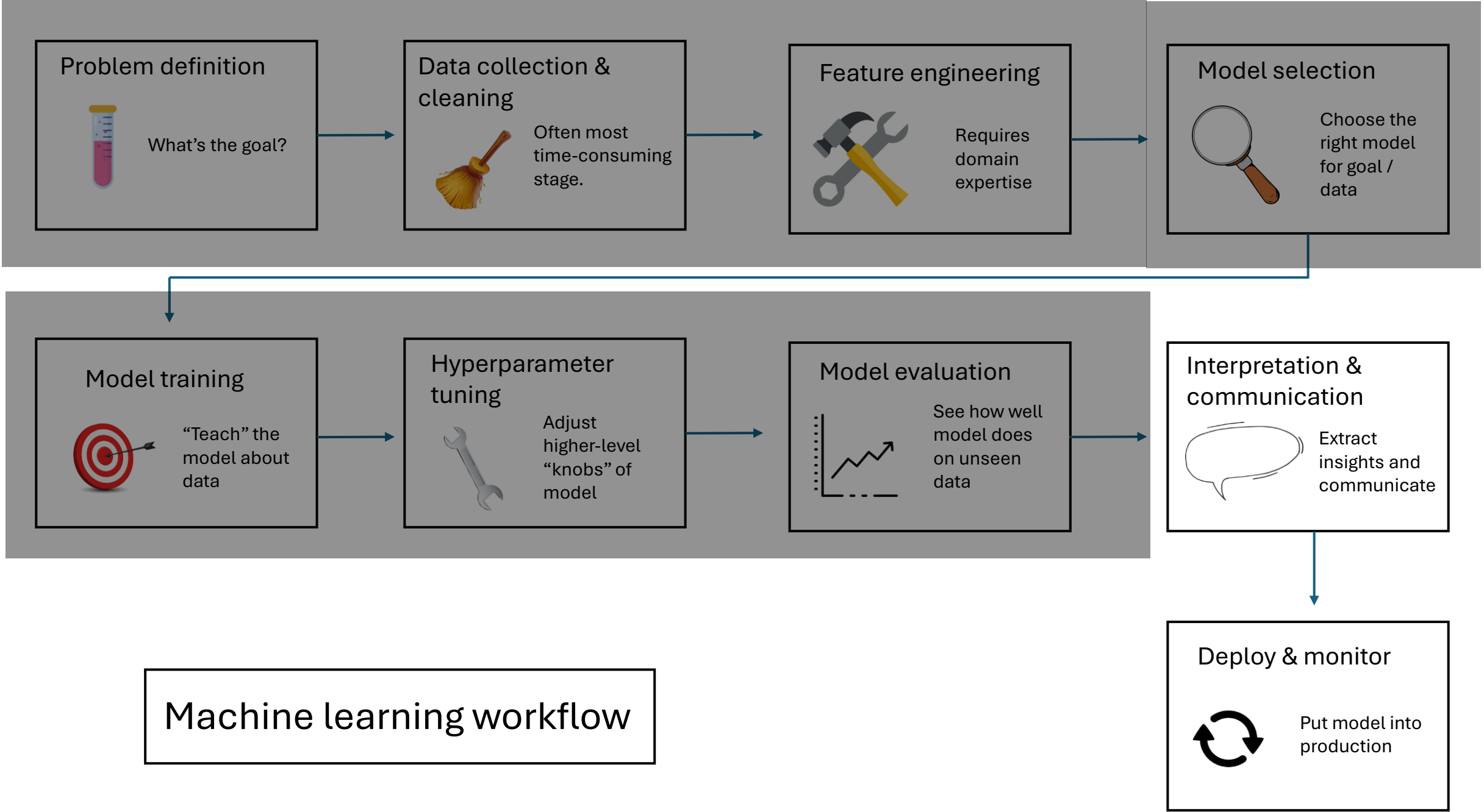
y



Validation target

Generalize beyond previous experience





Machine learning in the real world:

The **great** and the **concerning**

The **concerning:** Why the internet is the way it is

The image is a composite of three elements illustrating internet-related concerns:

- Top Left:** A screenshot of a CNN news article titled "Mike Waltz pushed out of Trump administration". The article features a photo of Mike Waltz and discusses his removal from the National Security Council. Below the article, there are "Live Updates" and "Trending" sections.
- Top Right:** A screenshot of a browser's domain list. The list shows various domains, with "amazon-adsystem.com" and its subdomains highlighted in a yellow box. This suggests a focus on advertising or tracking domains.
- Bottom:** A screenshot of a blocked page notification. It states "Blocked on this page 131 (11%)" and "Domains connected 13 out of 37". It also mentions "Blocked since install 71,031 (5%)". The notification is overlaid on a background that includes a large Google logo and an Amazon logo.

The **concerning:** Why the internet



Timestamp	Data Type	Value / Behavior	Source
2025-04-30 14:45	Off-Facebook Browsing	Read article "How to tell if you're burned out" on WebMD	Facebook Pixel on
2025-04-30 10:22	Location Check-in	Phone was at Planned Parenthood for 42 minutes	Location data from Instagram/Meta S
2025-04-29 23:09	Drafted Message	Typed but never sent "I think I might be depressed" on Messenger	Message draft log tracked)
2025-04-29 21:04	Group Browsing	Viewed private Facebook group: "ADHD support for late-diagnosed adults"	Logged user beha groups
2025-04-29 18:33	Third-party App Usage	Played 18 mins of "Calm" meditation app	Facebook SDK en
2025-04-29 17:47	Hover Behavior	Hovered over ex's Instagram profile pic for 5.4 seconds	Eye-catching beh Instagram
2025-04-28 16:12	Facial Analysis	Flagged low mood based on facial expression in selfie	Auto-tagging algo Facebook app
2025-04-28 14:23	Typing Speed Pattern	Detected slowed typing over past 5 days	Behavioral biome apps
2025-04-27 20:04	Audio Background Data	Detected raised voices and crying in room while app was open	Ambient sound an (undisclosed use)
2025-04-26 12:34	Contact Linking	Synced phone contacts showed frequent calls to mental health hotline	Contact + call me WhatsApp/Messenger



Welcome to Pinterest

Email

Password

Forgot your password?

Log in

OR

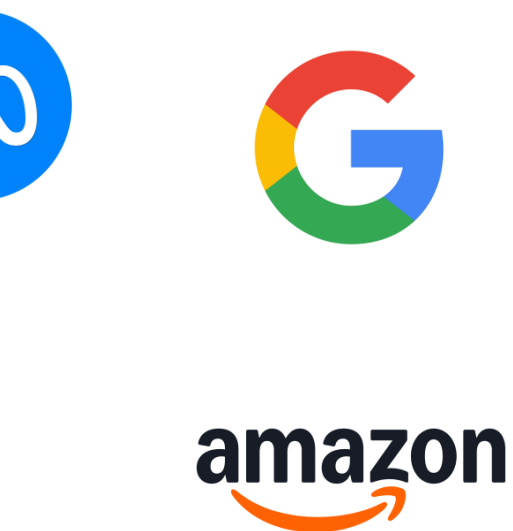
 Continue with Facebook

 Continue with Google

By continuing, you agree to Pinterest's [Terms of Service](#) and acknowledge you've read our [Privacy Policy](#). [Notice at collection](#).

Not on Pinterest yet? [Sign up](#)

Are you a business? [Get started here!](#)



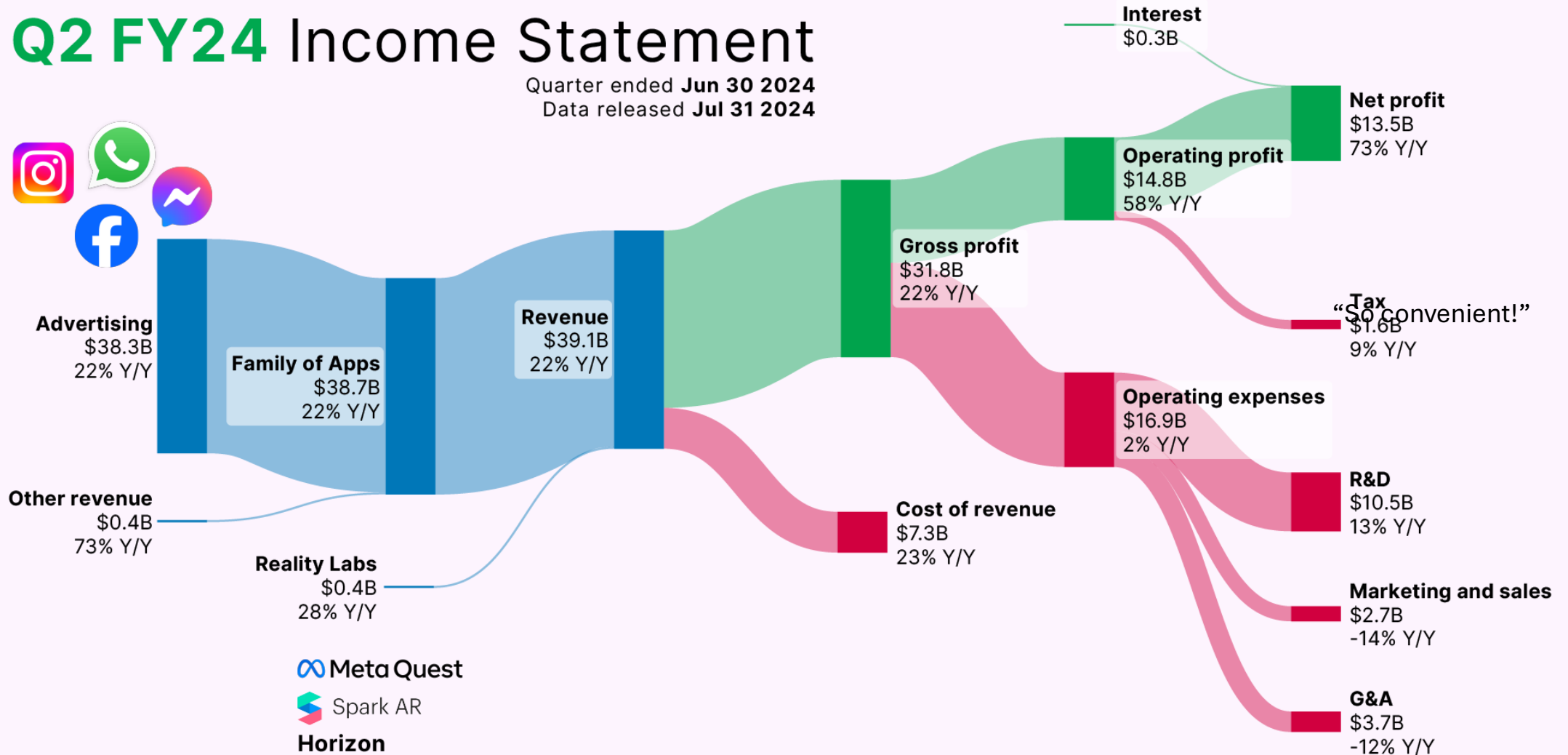
Time
2025-04-
2025-04-
2025-04-
2025-04-
2025-04-
2025-04-
2025-04-
2025-04-
2025-04-
2025-04-



Q2 FY24 Income Statement

Quarter ended Jun 30 2024
Data released Jul 31 2024

Source **Meta investor relations**
Created with **SankeyArt.com**





Marketing

Data brokers collect employment history, social media interactions, and financial records.
Many job applicants have shadow profiles without knowing—aggregated from third-party sources like credit reports and online résumés.
Employers purchase candidate profiles from brokers like Acxiom, Experian, and Equifax.

Public rec

Location

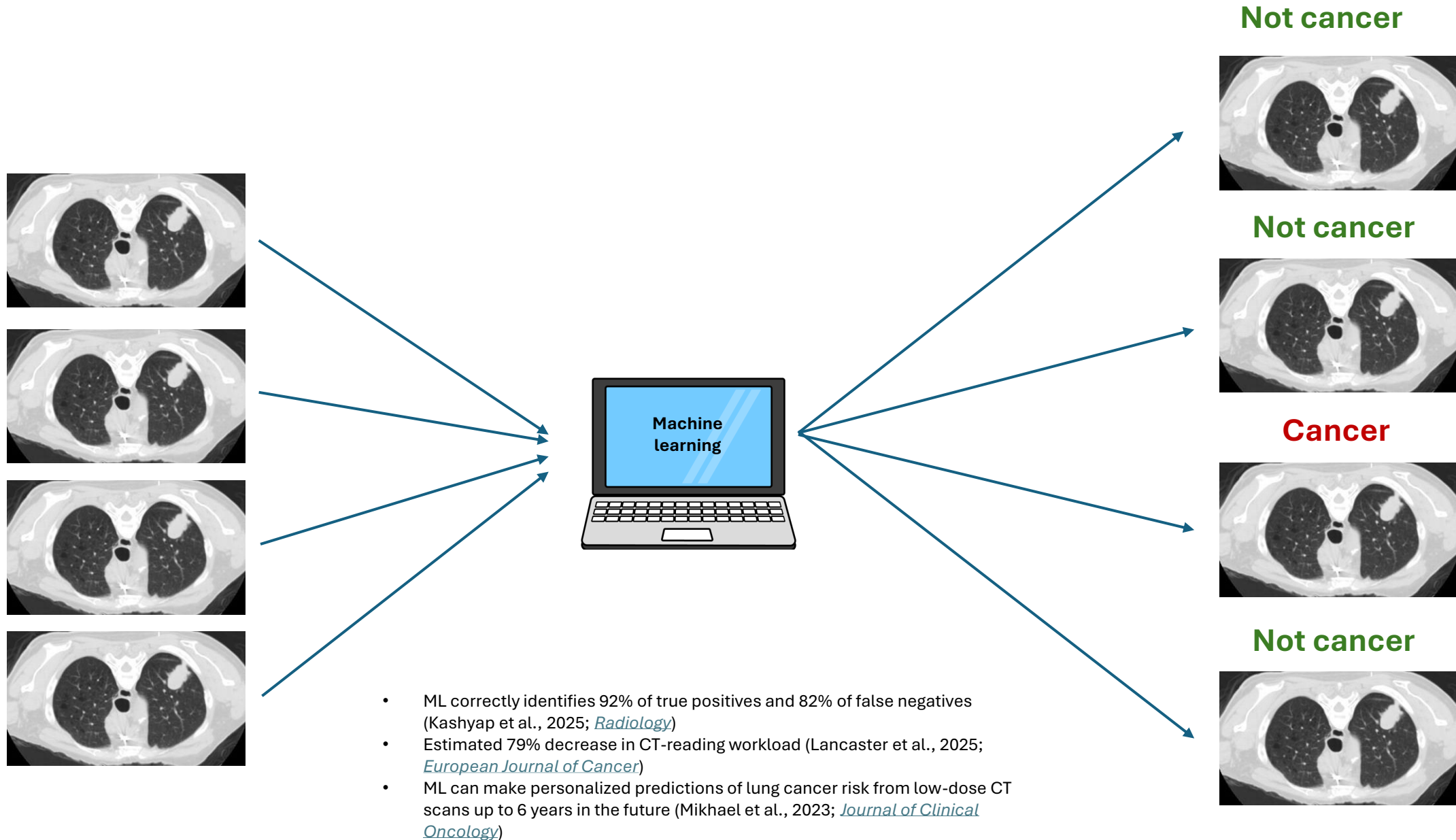
Kuempel, 2016

Northwestern Journal of International Law & Business

General app
data

Data brokers

Machine learning in the real world: The great



Thanks!